

Analysis of Multilayer Perceptron Architectures for Classifying Non-Linearly Separable Data

Problem Statement

Non-linearly separable datasets present a challenge to simple neural network models. In this experiment, you are required to analyze the limitations of a basic Perceptron and study how architectural depth and non-linear activation functions improve the learning capability of Multilayer Perceptrons (MLPs).

Objectives

Part A: Limitations of Simple Neural Architectures

1. Implement a basic neural network with:
 - One hidden layer
 - One output layer
 - Hard-limiting (step) activation function
2. Train the model using a non-linearly separable dataset such as:
 - make_circles dataset from scikit-learn
 - OR**
 - A custom dataset with overlapping class boundaries.
3. Evaluate the model using:
 - Classification accuracy
 - Decision boundary visualization
4. Demonstrate that the basic Perceptron fails to correctly classify non-linearly separable data.

Part B: Improving Performance Using Advanced MLP Architectures

Step 1: Increasing Network Depth

1. Design and train MLP models with:
 - Two hidden layers
 - Three hidden layers
2. Compare their classification performance with the baseline model.

Step 2: Effect of Non-Linear Activation Functions

1. Replace the hard-limiting activation function with:
 - Sigmoid activation
 - ReLU activation
2. Train and evaluate each model.

Step 3: Convergence Analysis

1. Analyze and compare Sigmoid and ReLU based models with respect to:
 - Training convergence speed
 - Number of epochs required for stable convergence
 - Gradient behavior during training

Part C: Comparative Evaluation

1. Compare all implemented models based on:
 - Classification accuracy
 - Loss curves
 - Decision boundary plots
2. Provide observations on:
 - The effect of network depth on learning non-linear patterns
 - The role of activation functions in learning efficiency
 - The relationship between model complexity and convergence behavior

Deliverables:

Python code for all the tasks mentioned above. Plots showing accuracy trends and decision boundaries. A report summarizing your analysis, observations, and conclusions.